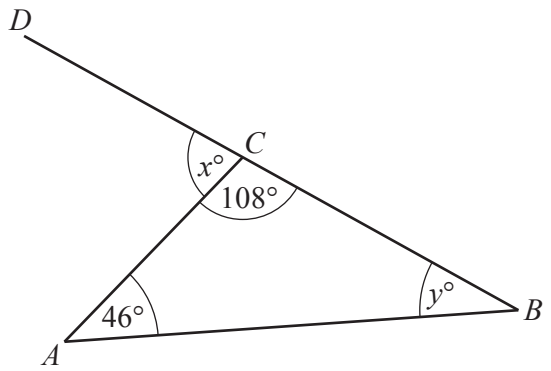
0580_s23_qp_33_Q7

7 **(a)**



NOT TO SCALE

The diagram shows a triangle *ABC* and a straight line *BCD*.

(i) Angle $ACB = 108°$.

Write down the mathematical name for this type of angle.

..... [1]

(ii) Work out the value of x.

$x =$ ..... [1]

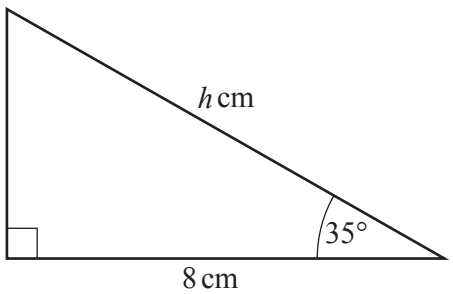
(iii) Work out the value of y.

$y =$ ..... [1]

(b) Show that the mean of the angles in any triangle is 60°.

[1]

(c)



NOT TO SCALE

The diagram shows a right-angled triangle.

Calculate the value of h.

$h =$ ..... [3]

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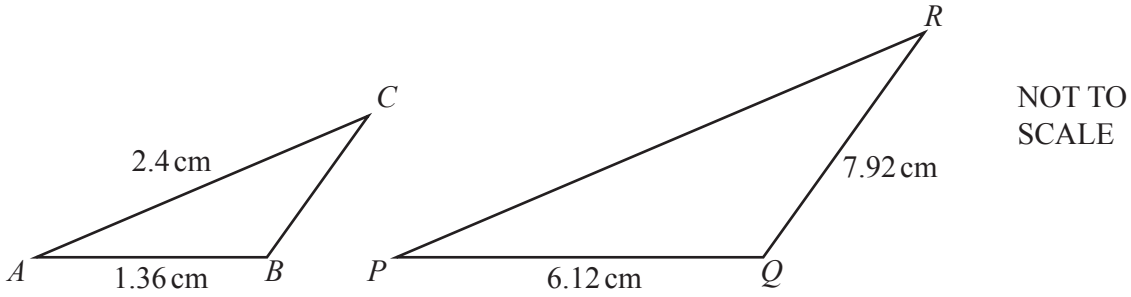
(d)



Triangle *ABC* is similar to triangle *PQR*.

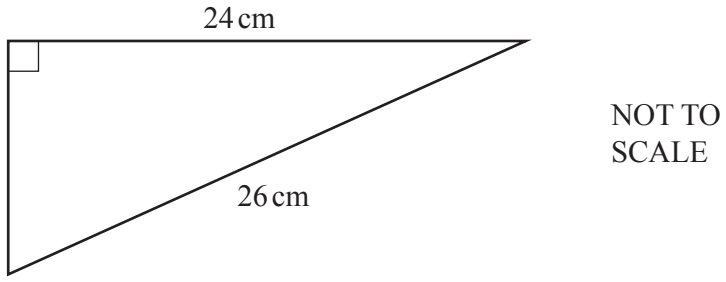
(i) Calculate *PR*.

PR = ... cm [2]

(ii) Calculate *BC*.

BC = ... cm [2]

(e)



The diagram shows a right-angled triangle.

Calculate the perimeter of this triangle.

..... cm [4]